MATH 350-2 Advanced Calculus

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Outline

1. Real Analysis Lecture 19
   - Uniform continuity
   - Differentiation

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Uniform continuity

Definition

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. We say that a function $f : A \to T$ is **uniformly continuous** on A if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_S(x, y) < \delta \Rightarrow d_T(f(x), f(y)) < \epsilon$$

for all $x, y \in A$.

- f continuous on A means f is continuous at every $a \in A$
- δ can depend on ϵ and a (not on x)
- with uniform continuity, δ can't depend on a ...

Examples

- x is uniformly continuous on $\mathbb{R}$
- x^2 is NOT uniformly continuous on $\mathbb{R}$
- $1/x$ is continuous on $(0, 1)$
- $1/x$ is NOT uniformly continuous on $(0, 1)$

Relationship with compactness

Theorem (Heine)

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$ be compact. Suppose that $f : A \to T$ is continuous on A, then f is uniformly continuous on A.

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Differentiability

Definition

A real-valued function $f(x)$ defined on an open interval (a, b) is **differentiable** at $c \in (a, b)$ if

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} \text{ exists.}$$

In this case the limit is called the derivative of f at c, and denote by $f'(c)$.

Chord function

Theorem (Helper Theorem)

Let f be a real-valued function on (a, b). Then f is differentiable at $c \in (a, b)$ if and only if there exists a function f^* on (a, b) which is continuous at c and satisfies

$$f(x) - f(c) = (x - c)f^*(x).$$

Chord function

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The function $f^*(x)$ is the **chord slope function**.

Chord function

Proof.

Chord function

Proof.

Suppose $f(x)$ is differentiable at c.

Chord function

Proof.

Suppose $f(x)$ is differentiable at c. Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

Chord function

Proof.

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Then $\lim_{x\to c} f^*(x) = f'(c)$, so f^* is continuous at c.

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Conversely, suppose f^* exists and is continuous at c.

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Conversely, suppose f^* exists and is continuous at c.
Then

$$\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = \lim_{x\to c} f^*(x) = f^*(c).$$

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Proof.

Suppose $f(x)$ is differentiable at c. Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

Then $\lim_{x\to c} f^*(x) = f'(c)$, so f^* is continuous at c.
Conversely, suppose f^* exists and is continuous at c.
Then

$$\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = \lim_{x\to c} f^*(x) = f^*(c).$$

In particular f is differentiable at c and $f'(x) = f^*(c)$. □

Differentiable implies continuous

Theorem

Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.
Then f is continuous at c.

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Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.
Then f is continuous at c.

Proof.

By the Helper Theorem, there exists a function f^* on (a, b) which is continuous at c with

$$f(x) = (x - c)f^*(x) + f(c).$$

Differentiable implies continuous

Theorem

Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.
Then f is continuous at c.

Proof.

By the Helper Theorem, there exists a function f^* on (a, b) which is continuous at c with

$$f(x) = (x - c)f^*(x) + f(c).$$

Since $(x - c)$ and f^* are both continuous at c, so is $f(x)$. □

Additivity of derivatives

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$.

$$(f \pm g)'(c) = f'(c) \pm g'(c)$$

Additivity of derivatives

Proof.

Additivity of derivatives

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Additivity of derivatives

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

Additivity of derivatives

Proof.

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Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

Since f^* and g^* are continuous at c, so is $f^* \pm g^*$.

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Proof.

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Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

Since f^* and g^* are continuous at c, so is $f^* \pm g^*$.
The Helper Theorem implies that $g \pm f$ is differentiable and

$$(f \pm g)'(c) = f'(c) \pm g'(c)$$



Product rule

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$.

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c)$$

Product rule

Proof.

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By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Product rule

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By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$\begin{aligned}(fg)(x) - (fg)(c) &= f(x)g(x) - f(c)g(c) \\ &= f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c) \\ &= f(x)(g(x) - g(c)) + (f(x) - f(c))g(c) \\ &= (x - c)[f(x)g^*(x) + f^*(x)g(c)]\end{aligned}$$

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By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

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Helper Theorem says:

$$(fg)'(c) = f(c)g^*(c) + f^*(c)g(c) = f(c)g'(c) + f'(c)g(c).$$

Quotient rule

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$ with $g(c) \neq 0$.

$$(f/g)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$$

Quotient rule

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Quotient rule

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

$$\begin{aligned}
(f/g)(x) - (f/g)(c) &= \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)} \\
&= \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)} \\
&= \frac{(f(x) - f(c))g(c) - f(c)(g(x) - g(c))}{g(x)g(c)} \\
&= (x - c)\frac{f^*(x)g(c) - f(c)g^*(x)}{g(x)g(c)}
\end{aligned}$$

Quotient rule

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

$$\begin{aligned}(f/g)(x) - (f/g)(c) &= \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)} \\ &= \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)} \\ &= \frac{(f(x) - f(c))g(c) - f(c)(g(x) - g(c))}{g(x)g(c)} \\ &= (x - c)\frac{f^*(x)g(c) - f(c)g^*(x)}{g(x)g(c)}\end{aligned}$$

Helper Theorem says:

$$(f/g)'(c) = \frac{f^*(c)g(c) - f(c)g^*(c)}{g(c)^2} = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}.$$

Chain rule

Theorem

Let f be a real function on an interval $I = (a, b)$, and let g be a real function on $f(I)$, and consider the composition

$$h = g \circ f : (a, b) \to \mathbb{R}.$$

Suppose $c \in (a, b)$ and $f(c)$ is an interior point of $f(I)$. If f is differentiable at c and g is differentiable at $f(c)$, then h is differentiable at c and

$$h'(c) = (g \circ f)'(c) = g'(f(c))f'(c).$$